Exp 4b – Rolling for Compound Initiative

INTRODUCTION

This activity is inspired by *Dungeons & Dragons*, a fantasy role-playing game where players take on the roles of adventurers—wizards, warriors, rogues, and more—who work together to explore magical worlds, solve puzzles, and overcome challenges using teamwork and creativity.

In our version, **you and your party are alchemists** on a quest to master the language of chemistry. Instead of casting spells or battling dragons, you'll be rolling 20-sided dice to generate mystery compounds, then using your naming skills to identify them correctly. Along the way, you'll encounter magical creatures, ancient scrolls, and arcane challenges—all themed around chemical nomenclature.

You don't need any experience with *Dungeons & Dragons* to succeed—just bring your chemistry knowledge, your sense of adventure, and a little imagination. May your compounds be correct and your dice rolls be ever in your favor.

BACKGROUND

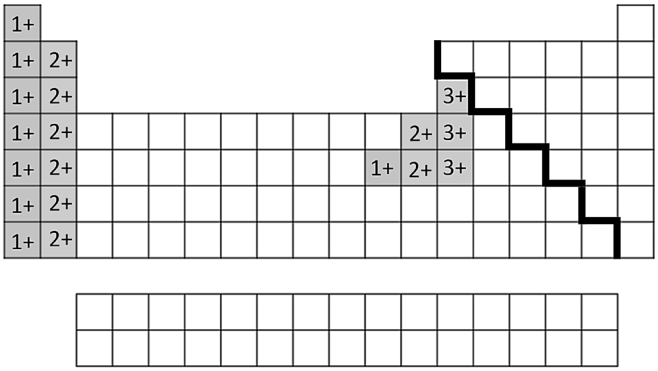
Every type of chemical compound has its own set of nomenclature rules. This activity is designed to help you practice identifying compound types and applying the rules to the different types of compounds, or deciphering the name of a compound to achieve the correct chemical formula.

Step 1: Identify the type of Compound

- **Ionic Compound:** Contains a metal + nonmetal, or includes polyatomic ions that have combined to be neutral overall (i.e. $NaCl, Ca(NO_3)_2$)
  - Metal elements form positively charged ions called ***cations***
  - Nonmetal elements form negatively charged ions called ***anions***
  - Opposite charges (cations and anions) attract and form **ionic bonds**.
  - Ions can be:
    - Monatomic (single element): Na^+, Cl^-, Fe^{3+}, etc.
    - Polyatomic (many elements): $NH_4^+, NO_3^-, C_2H_3O_2^-$, etc.
- **Molecular Compound:** Contains only nonmetals (i.e. CO_2, N_2O_4)
  - Will be neutral (not charged)
  - Nonmetal elements form covalent bonds by sharing electrons.
- **Acid:** Chemical formula starts with H. Hydrogen will be bonded with a nonmetal anion or a polyatomic anion.
  - When dissolved in water (aqueous solution), H^+ions separate from the nonmetal anion, or polyatomic anion.

Step 2: Naming Rules

- **Ionic Compounds:**
  - **Type I Ionic Compounds:** Metal cation only forms one ion, as shown below
    The ammonium ion, NH_4^+, also forms Type I ionic compounds since it is always a 1+ charge.



    - Name with monatomic anion: [Metal/Cation] + [Nonmetal/Anion]-ide
      - **Example:** $CaCl_2$
        $\boldsymbol{Ca^{2+}} = \boldsymbol{calcium}\ ion$ $\boldsymbol{Cl^-} = \boldsymbol{chloride}\ ion$

$CaCl_2$ = calcium chloride

One calcium cation and two chloride anions are needed to form a neutral compound.

    - Name with polyatomic ions: keep their name as-is: [Metal//Cation] + [Anion]
      - Suffix for anion is typically **-ate** or **-ite**
      - **Example: $(NH_4)_2CO_3$**
        $\boldsymbol{NH_4^+} = \boldsymbol{ammonium}\ ion$ $\boldsymbol{CO_3^{2-}} = \boldsymbol{carbonate}\ ion$

$(NH_4)_2CO_3$ = ammonium carbonate

Two ammonium cations and one carbonate ion are needed to form a neutral compound. When more than one of a polyatomic ion are needed, parentheses are used around the polyatomic ion's formula.

- **Type II Ionic Compounds:** Metal can form more than one ion. Any metal that isn't Type I.
  - Name with monatomic anion: [Metal](Roman numeral for charge) + [Nonmetal]-ide
    - **Example:** $FeCl_3$
      $\boldsymbol{Fe^{3+}} = \boldsymbol{iron(III)}\ ion$ $\boldsymbol{Cl^-} = \boldsymbol{chloride}\ ion$

$FeCl_3$ = iron(III) chloride

  - Name with polyatomic ions: keep their name as-is:
    [Metal](Roman numeral for charge) + [Anion]
    - **Example: $\boldsymbol{Cu(NO_3)_2}$**
      $\boldsymbol{Cu^{2+}} = \boldsymbol{copper(II)}\ ion$ $\boldsymbol{NO_3^-} = \boldsymbol{nitrate}\ ion$

$Cu(NO_3)_2$ = copper(II) nitrate

- **Molecular Compounds**
    - Use Greek prefixes to specify the number of each atom (quantity of each atom)

How many	1	2	3	4	5	6	7	8	9	10
Prefix	Mono-	Di-	Tri-	Tetra-	Penta-	Hexa-	Hepta-	Octa-	Nona-	Deca-

    - The first element never gets "mono-" prefix if there is only one atom
    - The second element's ending gets changed to **-ide**
    - **Examples:**
        - CO_2 = carbon **di**ox**ide**
        - N_2O_4 = **di**nitrogen **tetra**ox**ide**

- **Acids**
    - **Binary Acids:** H^+ + nonmetal (no oxygen)
        - **Name:** hydro-[root]-ic acid
        - **Examples:** HCl = hydro**chlor**ic acid

    - **Oxyacids:** H^+ + polyatomic anion with oxygen
        - **Name (polyatomic anion ends with -ate) : [root]-ic acid**
            - **Example:** $\boldsymbol{H_2SO_4} \rightarrow \boldsymbol{SO_4^{2-}} = sulf\boldsymbol{ate}\ ion$
              $\boldsymbol{H_2SO_4}$ = sulfur**ic acid**
        - **Name (polyatomic anion ends with -ite): [root]-ous acid**
            - **Example:** $\boldsymbol{H_2SO_3} \rightarrow \boldsymbol{SO_3^{2-}} = sulf\boldsymbol{ite}\ ion$
              $\boldsymbol{H_2SO_3}$ = sulfur**ous acid**

ALLOWED SCROLLS AND ARTIFACTS (SUPPLY LIST)

- Two 20-sided dice (2d20)
- Two 10-sided dice (2d10)
- Tray (to contain dice)
- Other instructor-approved resources (such as textbook, class notes, etc.)

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

Each die roll combination should be unique. If a duplicate dice roll combination is obtained, roll again.

PHASE 1: IONIC COMPOUNDS

	Dice #1 Roll	Dice #2 Roll	Compound Chemical Formula	Cation Formula	Type I or Type II?	Anion Formula	Compound Name
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							

PHASE 2: ACIDS

	Dice #1 Roll	Compound Chemical Formula	Cation Formula	Anion Formula	Binary Acid or Oxyacid?	Compound Name
13						
14						
15						
16						
17						
18						
19						
20						
21						

PHASE 3: MOLECULAR COMPOUNDS

	Dice #1 Roll	Dice #2 Roll	Compound Chemical Formula	How many of first element?	Prefix for quantity of first element?*	How many of second element?	Prefix for quantity of second element?*	Compound Name
22								
23								
24								
25								
26								
27								
28								
29								

* Prefixes associated with numerical quantity (i.e. mono, di, tri, tetra, etc.)

FINAL BATTLE

Chemical Formula	Type of Compound	Name

FINAL BATTLE SCRATCH WORK